

Satvik Kaushal

Forward Deployed Engineer · Enterprise AI Development & Deployment

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SUMMARY

Senior engineer with 7+ years building and deploying production software's, with deep recent focus on AI and Agentic systems inside complex & regulated enterprise environments. I've built and shipped production MCP servers, agentic pipelines, and LLM integrations - working directly alongside Business stakeholders to translate real workflows into running software. I own the full path from scoping a use case with end users, to building the integration, to breaking through the enterprise wall - SSO/OAuth, locked-down networks, cross-account cloud, and security review - to driving adoption. Comfortable in ambiguity, customer-facing, and biased toward shipping measurable outcomes.

CORE SKILLS

- **AI/Agentic:** Production LLM applications, Agentic workflows, Multi Agents/Tools use, MCP Clients & Servers Design, Context Engineering, RAG, Azure OpenAI, LangChain, LangGraph, RAGAS, DeepEval, LangFuse, Helicone, Chunking, Reranking, Embeddings, Vector-less RAG, Deep Agents, LLM & MCP Gateways.
- **Enterprise integration:** Azure Entra ID, OAuth 2.0 / OIDC, MSAL, SSO, API Gateways, third-party API integration, Data-residency and Security-review navigation
- **Cloud & infra:** AWS (Lambda, ECS Fargate, EKS, Glue, Step Functions, Athena, S3, API Gateway, CloudFront, DynamoDB, Event Bridge, CloudFormation, SQS), Docker, Kubernetes, OpenShift, ALB, Terraform, CI/CD.
- **Languages & Full stack:** Python, JavaScript / TypeScript, Node.js, React JS, Next.js, Fast API, GraphQL, REST API, PostgreSQL, MongoDB, Redis.

EXPERIENCE

Principal Software Engineer - Eli Lilly and Company

Aug 2022 - Present

Bengaluru, India · Lilly Research Laboratories (LRL)

- **Built and deployed a production MCP server** exposing multiple agentic tools over enterprise lab instrument data (DynamoDB/S3) spanning multiple R&D labs and instrument types, taking an AI capability from sandbox to production in a regulated pharma environment using FastMCP on AWS ECS Fargate behind an ALB & Secured Enterprise MCP Gateway.
- **Led AIDA, an AI Drafting Assistant**, using a RAG pipeline - semantic chunking, vector embeddings with top-K retrieval, score-threshold filtering, and reranking - with multi-step reasoning and LLM-driven document generation. Instrumented quality via RAGAS evaluation framework and LangFuse observability. Cut Japanese PSR document turnaround from 3 - 4 weeks to 2 - 3 days. Managed a team of four full-stack engineers.
- **Implemented Tools-level RBAC** by validating Microsoft Entra ID app roles from the OAuth access token, gating MCP tool access per user role so lab scientists, IT, and admins each see only authorized tools/data - meeting least-privilege requirements for a GxP - regulated environment.
- **Broke through the enterprise integration wall** - Implemented Azure AD OAuth (Authorization Code flow, token caching, redirect-URI and AADSTS troubleshooting), and resolved Zscaler / Artifactory connectivity, cross-account AWS Private Link, and Kubernetes deployment failures that stood between a working demo and a real deployment.
- **Designed and shipped a custom Auth/Authorization layer** using AWS Lambda Edge, WAF, MSAL, and CloudFront, and optimized an AWS Glue pipeline from 2 - 3 hours to 15 minutes through parallel processing and database optimizations.

- **Implemented the data module in CAMP** (Clinical Account Management & Processing), building pipelines handling 1M+ records daily with AWS Glue, Step Functions, Lambda, Event Bridge, and CloudFormation; onboarded VEEVA IWRS, Fircrest within six months.
- **Improved system performance by 30%** via lazy loading, GraphQL, and REST optimization, and built reusable React / TypeScript / Python components powering real-time Monte Carlo simulations in PPM.
- **Hardened orchestration reliability** with automated retries, dead-letter handling, and detailed logging/alerting across Glue and Lambda workflows, plus AWS SNS for real-time job notifications.
- **Delivered end-to-end scientific workflow automation** by integrating three platforms - L7 (workflow automation), Benchling (ELN/experiment data), and TetraScience TDP & the Data Capture Platform (instrument data) - enabling automated data flow from instrument to system of record across two R&D organizations (SMDD and BRD).
- **Contributed to the Lilly Heritage Hall 3D experience** on Oculus devices, presented the initiative to the CEO.

Technology Consultant - Deloitte

Dec 2020 - Aug 2022

Hyderabad, India

- **Built and optimized APIs** (Node.js, MongoDB, React) for a Test Automation Suite in a microservices architecture, improving modularity, scalability, and fault isolation.
- **Streamlined CI/CD on OpenShift** - automated build, test, and deployment, reducing release times and manual intervention.

Junior Associate Consultant - Rythmos Services India

May 2019 - Dec 2020

Hyderabad, India

- **Delivered migration services** for a large-scale loyalty application (Node.js, Java, MongoDB), containerized environments with Docker / Docker Compose, and drove test coverage via Jest / Mocha and TDD practices.

PROJECTS

- **Memward** - Self-hosted memory layer that ingests conversation context from Claude Code, Copilot, and Claude Desktop into a single owned store, giving assistants persistent cross-tool recall. Stack: FastAPI · Supabase · AWS
- **Odin** - a conversational CLI AI Agent with a tool-use loop engineering framework, persistent cross-session memory, live web search, and system tools. Currently kept local while hardening auth and transport.
- **Mcp-rc-audit** - an open-source conformance auditor for MCP servers, a Python CLI pairing a static scanner with a live probe to verify a server actually behaves as the spec claims.

RECOGNITIONS

- Lilly Innovator Award winner for the AIDA project (2025)
- Finalist at the Tech at Lilly CIDO Awards, for high-impact contributions across CAMP and AIDA
- Multiple RISE Awards for technical excellence and end-to-end ownership of high-impact engineering deliverables.

EDUCATION

B Tech, Electronics & Communication Engineering

April - 2019

Raghu Engineering College (JNTU Kakinada), Visakhapatnam

LANGUAGES

English (Proficient) · Hindi (Native) · Telugu (Proficient)